

Lecture 3 – Sufficiency

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3.1 Data reduction

Recall the statistical framework from Lecture 1.

- A **statistical model** is a family of distributions

$$\{f_{\theta}(x) : \theta \in \Theta\}.$$

The unknown parameter θ indexes the possible distributions. Suppose x lives in $\mathcal{X}$ (e.g. $\mathcal{X} = \mathbb{R}$ for a normal distribution and $\mathcal{X} = \{0, 1\}$ for a Bernoulli). We observe a realization $\underline{x} = (x_1, \dots, x_n) \in \mathcal{X}^n$ of a random sample

$$\underline{X} = (X_1, \dots, X_n), \quad X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f_{\theta}.$$

By independence, the joint pmf or pdf of the sample is

$$f_{\underline{X}, \theta}(\underline{x}) = \prod_{i=1}^n f_{\theta}(x_i).$$

- The goal of **statistical inference** is to learn about the unknown θ from the observed sample. For example, we may construct a point estimate, test a hypothesis, or form a confidence interval. In every case, our conclusion is a function of the data $\underline{X}$.

To make inference more manageable, we first try to discard extraneous information. We replace the full sample by a statistic $T(\underline{X})$ that summarizes the part of the data relevant to θ . This is called **data reduction**.

Example 1 (Bernoulli motivation). Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p),$$

where p is the unknown probability of success. Suppose $n = 3$ and we observed

$$\underline{x} = (1, 1, 0).$$

Intuitively, this is more information than we need: it is enough to know that two of the three observations are successes to learn about p . This information is recorded by

$$T(\underline{x}) = \bar{x} = \frac{1}{3} \sum_{i=1}^3 x_i = \frac{2}{3}.$$

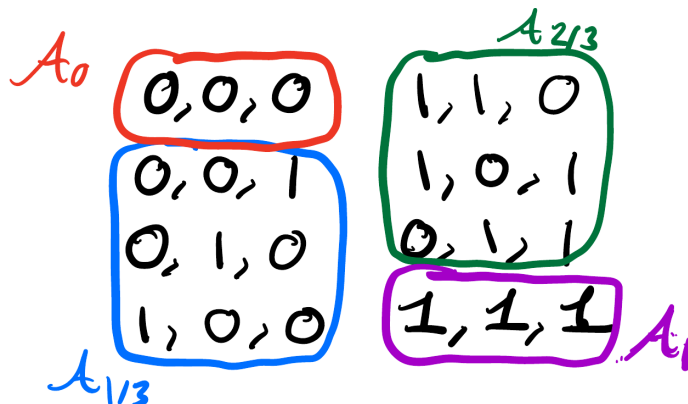
The samples $(1, 1, 0)$ and $(0, 1, 1)$ should lead to the same conclusion about p .

More generally, any statistic T partitions the sample space. For every possible value t of T , define the **fiber**

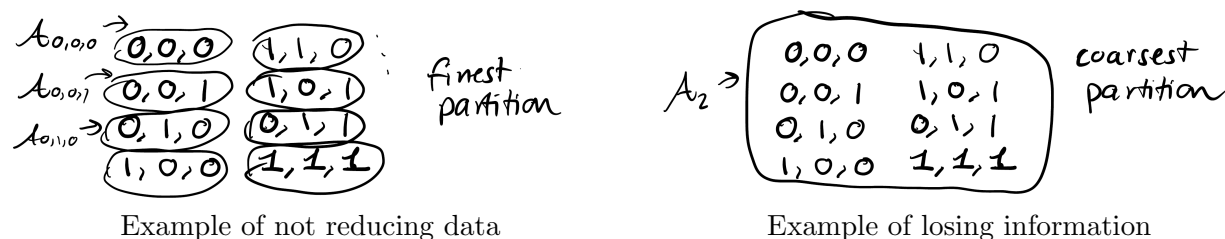
$$A_t = \{\underline{x} \in \mathcal{X}^n : T(\underline{x}) = t\}.$$

The statistic value t tells us which fiber A_t contains the observed sample, but not which particular point in that fiber was observed.

For $n = 3$ Bernoulli observations and $T(\underline{x}) = \bar{x}$, the partition is



The finer the partition, the less data reduction we perform. For example, $T(\underline{x}) = \underline{x}$ gives singleton fibers and performs no reduction (below, left). The coarser the partition, the more reduction we perform, and we risk losing information. For example, a constant statistic such as $T(\underline{x}) = 2$ puts every sample point in one fiber and throws away all the information (below, right).



Example of not reducing data

Example of losing information

Our goal is to choose a statistic in the middle between these two: it should *reduce* the data *without losing information* about θ .

3.2 Sufficient Statistics

The key intuition is that we may discard any part of the data whose distribution is unaffected by θ . A sufficient statistic retains all of the θ -dependence in the sample; once its value is known, the remaining variation in the data carries no further information about θ .

Definition 1 (Sufficient statistic). A statistic $T(\underline{X})$ is **sufficient** for θ if, for every possible value t , the conditional distribution of $\underline{X}$ given $T(\underline{X}) = t$ does not depend on θ .

To understand the definition, note that we can think of the full sample as being generated in two stages:

1. Draw t from the distribution of $T(\underline{X})$.
2. Given $T(\underline{X}) = t$, draw $\underline{X}$ from its conditional distribution.

If T is sufficient, then θ appears only in the first stage, whereas the second stage carries no extra information about θ .

The two-stage structure is reflected in the pdf/pmf of $\underline{X}$. Let's understand this in the discrete case. We first compute the conditional distribution of $\underline{X}$ given $T(\underline{X}) = t$. Fix a sample point $\underline{x}$ such that $t = T(\underline{x})$. Since

$$\{\underline{X} = \underline{x}\} \subseteq \{T(\underline{X}) = t\},$$

we have

$$\begin{aligned} \mathbb{P}_\theta(\underline{X} = \underline{x} \mid T(\underline{X}) = t) &= \frac{\mathbb{P}_\theta(\underline{X} = \underline{x}, T(\underline{X}) = t)}{\mathbb{P}_\theta(T(\underline{X}) = t)} \\ &= \frac{\mathbb{P}_\theta(\underline{X} = \underline{x})}{\mathbb{P}_\theta(T(\underline{X}) = t)}. \end{aligned}$$

Equivalently,

$$f_{\underline{X}, \theta}(\underline{x}) = \mathbb{P}_\theta(\underline{X} = \underline{x}) = \underbrace{\mathbb{P}_\theta(T(\underline{X}) = t)}_{\text{Stage 1: sample } T} \underbrace{\mathbb{P}_\theta(\underline{X} = \underline{x} \mid T(\underline{X}) = t)}_{\text{Stage 2: sample } \underline{X} \mid T} \quad (3.1)$$

The decomposition (3.1) is true whether or not T is sufficient. But if T is sufficient, then only the first factor in the product depends on θ . The second factor does not, so we could have written $\mathbb{P}(\underline{X} = \underline{x} \mid T(\underline{X}) = t)$ instead of $\mathbb{P}_\theta(\underline{X} = \underline{x} \mid T(\underline{X}) = t)$.

Example 2 (Sufficient statistic for Bernoulli sample). Let $X_1, X_2, X_3 \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ and $T(\underline{X}) = \bar{X}$. The distribution of T is

$$\begin{array}{c|cccc} t & 0 & \frac{1}{3} & \frac{2}{3} & 1 \\ \hline \mathbb{P}_p(T = t) & (1-p)^3 & 3p(1-p)^2 & 3p^2(1-p) & p^3. \end{array}$$

This distribution clearly depends on the unknown parameter p , so $T(\underline{X})$ carries information about p . To check whether $T(\underline{X})$ is sufficient, we need to show $\underline{X}$ given T does *not* carry information about p . Consider for example the case $T(\underline{X}) = 2/3$. Then $\underline{X}$ could equal $(1, 1, 0)$, $(1, 0, 1)$, or $(0, 1, 1)$. We have

$$\mathbb{P}_p(\underline{X} = (1, 1, 0) \mid T = 2/3) = \frac{p^2(1-p)}{3p^2(1-p)} = \frac{1}{3}.$$

The same calculation holds for the other two sample points, so the pmf of $\underline{X}$ given $T = 2/3$ is:

$$\begin{array}{c|ccc} \underline{x} & (1, 1, 0) & (1, 0, 1) & (0, 1, 1) \\ \hline \mathbb{P}_p(\underline{X} = \underline{x} \mid T = 2/3) & 1/3 & 1/3 & 1/3 \end{array}$$

Therefore, conditional on $T = 2/3$, the probabilities of the three possible orderings *do not depend* on p ! The same is true on every fiber (for each of $t = 0, 1/3, 2/3, 1$) so T is sufficient for p .

Example 3 (Insufficient statistic for Bernoulli sample). Let $X_1, X_2, X_3 \stackrel{\text{iid}}{\sim} \text{Ber}(p)$ and $T(\underline{X}) = X_1$. The distribution of T is

$$\begin{array}{c|cc} t & 0 & 1 \\ \hline \mathbb{P}_p(T = t) & 1-p & p \end{array}$$

This distribution clearly depends on the unknown parameter p , so $T(\underline{X})$ carries information about p . But $\underline{X} \mid T$ does too. For example, $\mathbb{P}_p(\underline{X} = (0, 1, 0) \mid T = 0) = \mathbb{P}_p(X_2 = 1, X_3 = 0) = p(1-p)$ depends on p . So $T(\underline{X}) = X_1$ is insufficient for p .

3.3 The Neyman–Fisher Factorization Theorem

The definition tells us how to verify whether a proposed statistic is sufficient, but it does not tell us how to find one. The following theorem gives a constructive criterion: start with the joint pmf or pdf and collect all dependence on θ through a statistic.

Theorem 1 (Neyman–Fisher Factorization Theorem). A statistic T is sufficient for θ in the statistical model $\{f_\theta(x) : \theta \in \Theta\}$ if and only if there exist nonnegative functions $g(t, x)$ and $h(x)$ such that the pdf/pmf $f_{\underline{X}, \theta}$ of the sample factorizes:

$$f_{\underline{X}, \theta}(\underline{x}) = g(T(\underline{x}), \theta) h(\underline{x})$$

for every $\underline{x}$ and θ , where h does not depend on θ .

Proof in the discrete case. First suppose that T is sufficient. For $t = T(\underline{x})$, write

$$\begin{aligned} f_{\underline{X}, \theta}(\underline{x}) &= \mathbb{P}_\theta(\underline{X} = \underline{x}) \\ &= \mathbb{P}_\theta(T(\underline{X}) = t) \mathbb{P}_\theta(\underline{X} = \underline{x} \mid T(\underline{X}) = t). \end{aligned}$$

Define

$$g(t, \theta) = \mathbb{P}_\theta(T(\underline{X}) = t)$$

and

$$h(\underline{x}) = \mathbb{P}_\theta(\underline{X} = \underline{x} \mid T(\underline{X}) = T(\underline{x})).$$

Because T is sufficient, the conditional probability defining h does not depend on θ . Hence the required factorization holds.

Conversely, suppose

$$f_{\underline{X}, \theta}(\underline{x}) = g(T(\underline{x}), \theta) h(\underline{x}).$$

For a value t with positive probability and any $\underline{x} \in A_t$,

$$\begin{aligned} \mathbb{P}_\theta(\underline{X} = \underline{x} \mid T(\underline{X}) = t) &= \frac{g(t, \theta) h(\underline{x})}{\sum_{\underline{y} \in A_t} g(t, \theta) h(\underline{y})} \\ &= \frac{h(\underline{x})}{\sum_{\underline{y} \in A_t} h(\underline{y})}. \end{aligned}$$

The last expression does not depend on θ , so T is sufficient. □

Example 4 (Bernoulli model). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$. The joint pmf is

$$\begin{aligned} f_{\underline{X}, p}(\underline{x}) &= \prod_{i=1}^n [p^{x_i} (1-p)^{1-x_i} \mathbb{1}_{\{x_i \in \{0,1\}\}}] \\ &= p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i} \mathbb{1}_{\{\underline{x} \in \{0,1\}^n\}}. \end{aligned}$$

Taking

$$T(\underline{X}) = \sum_{i=1}^n X_i,$$

we obtain the factorization

$$g(t, p) = p^t (1-p)^{n-t}, \quad h(\underline{x}) = \mathbb{1}_{\{\underline{x} \in \{0,1\}^n\}}.$$

Therefore T is sufficient for p . We can also verify this directly from the definition. Since $T \sim \text{Bin}(n, p)$, for any $\underline{x} \in \{0, 1\}^n$ with $\sum_i x_i = t$,

$$\begin{aligned}\mathbb{P}_p(\underline{X} = \underline{x} \mid T = t) &= \frac{p^t(1-p)^{n-t}}{\binom{n}{t}p^t(1-p)^{n-t}} \\ &= \frac{1}{\binom{n}{t}},\end{aligned}$$

which does not depend on p .

Example 5 (Normal model). Let

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \quad \theta = (\mu, \sigma^2),$$

so both μ and σ^2 are unknown. Then

$$\begin{aligned}f_{\underline{X}, \theta}(\underline{x}) &= (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right\} \\ &= (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2\right)\right\}.\end{aligned}$$

All dependence on the sample enters through

$$T(\underline{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2\right).$$

Indeed, with $t = (t_1, t_2)$, take

$$g(t, (\mu, \sigma^2)) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{t_2 - 2\mu t_1 + n\mu^2}{2\sigma^2}\right\}, \quad h(\underline{x}) = 1.$$

The factorization theorem shows that T is sufficient for (μ, σ^2) .

Example 6 (Uniform model). Let

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta), \quad \theta > 0.$$

Write $x_{(1)} = \min_i x_i$ and $x_{(n)} = \max_i x_i$. The joint density is

$$\begin{aligned}f_{\underline{X}, \theta}(\underline{x}) &= \prod_{i=1}^n \frac{1}{\theta} \mathbb{1}_{\{0 < x_i < \theta\}} \\ &= \theta^{-n} \mathbb{1}_{\{x_{(n)} < \theta\}} \mathbb{1}_{\{x_{(1)} > 0\}}.\end{aligned}$$

Thus, for

$$T(\underline{X}) = X_{(n)},$$

we may take

$$g(t, \theta) = \theta^{-n} \mathbb{1}_{\{t < \theta\}}, \quad h(\underline{x}) = \mathbb{1}_{\{x_{(1)} > 0\}}.$$

Therefore the sample maximum is sufficient for θ . Once the maximum is known, the locations of the remaining observations below it contain no additional information about θ .

3.4 Sufficiency and Further Data Reduction

The full sample $T^*(\underline{X}) = \underline{X}$ is always sufficient, but it performs no data reduction. More generally, adding information to a sufficient statistic preserves sufficiency.

Theorem 2. Suppose T is sufficient for θ and T^* is another statistic. If there is a function r such that

$$T = r(T^*),$$

then T^* is also sufficient for θ .

Remark 1. The reason we say T^* has added information about the sample compared to T is that knowing $T^* = t^*$ gives us the value of T (it is $r(t^*)$). On the other hand, knowing $T = t$ does not give us the value of T^* (it could be *any* value t^* such that $r(t^*) = t$, and we don't know which one). Thus T^* has more information.

Proof. Since T is sufficient, the factorization theorem gives

$$f_{\underline{X},\theta}(\underline{x}) = g(T(\underline{x}), \theta)h(\underline{x}).$$

Substituting $T = r(T^*)$ gives

$$f_{\underline{X},\theta}(\underline{x}) = g(r(T^*(\underline{x})), \theta)h(\underline{x}) = g^*(T^*(\underline{x}), \theta)h(\underline{x}),$$

where $g^*(t^*, \theta) = g(r(t^*), \theta)$. Another application of the factorization theorem shows that T^* is sufficient. $\square$

Example 7 (Several sufficient statistics for the normal model). For $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, we have shown that

$$T(\underline{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is sufficient. Each of the following more detailed statistics is also sufficient because T can be recovered from it.

First, let

$$T_1(\underline{X}) = \left(\bar{X}, \frac{1}{n} \sum_{i=1}^n X_i^2 \right).$$

Then

$$r_1(t_1, t_2) = (nt_1, nt_2).$$

Second, let

$$T_2(\underline{X}) = \left(\bar{X}, \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right).$$

Using

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2,$$

we recover T through

$$r_2(t_1, t_2) = (nt_1, n(t_2 + t_1^2)).$$

For $n \geq 5$, we may also add completely extraneous coordinates:

$$T_3(\underline{X}) = \left(\sum_i X_i, \sum_i X_i^2, \sum_i X_i^3, X_5 \right), \quad r_3(t_1, t_2, t_3, t_4) = (t_1, t_2).$$

Finally, the entire sample is sufficient:

$$T_4(\underline{X}) = \underline{X}, \quad r_4(\underline{x}) = \left(\sum_i x_i, \sum_i x_i^2 \right).$$

These examples emphasize that sufficiency alone does not identify the statistic that gives the greatest possible data reduction.